

JANUARY 2025

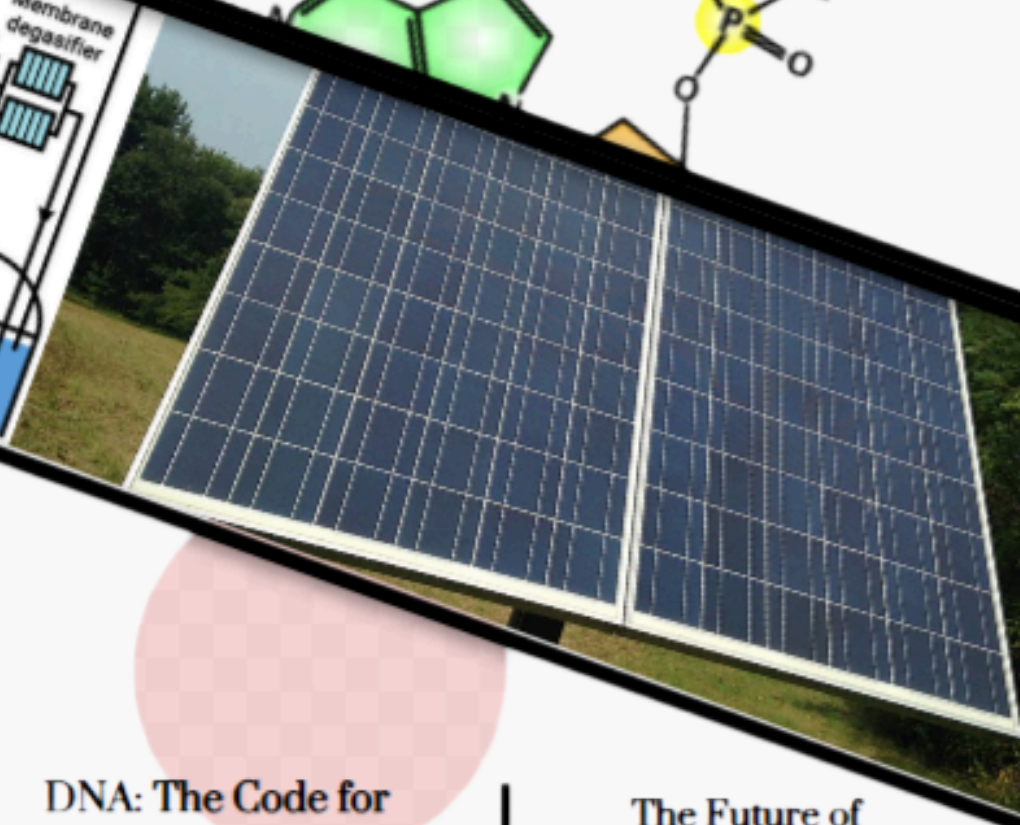
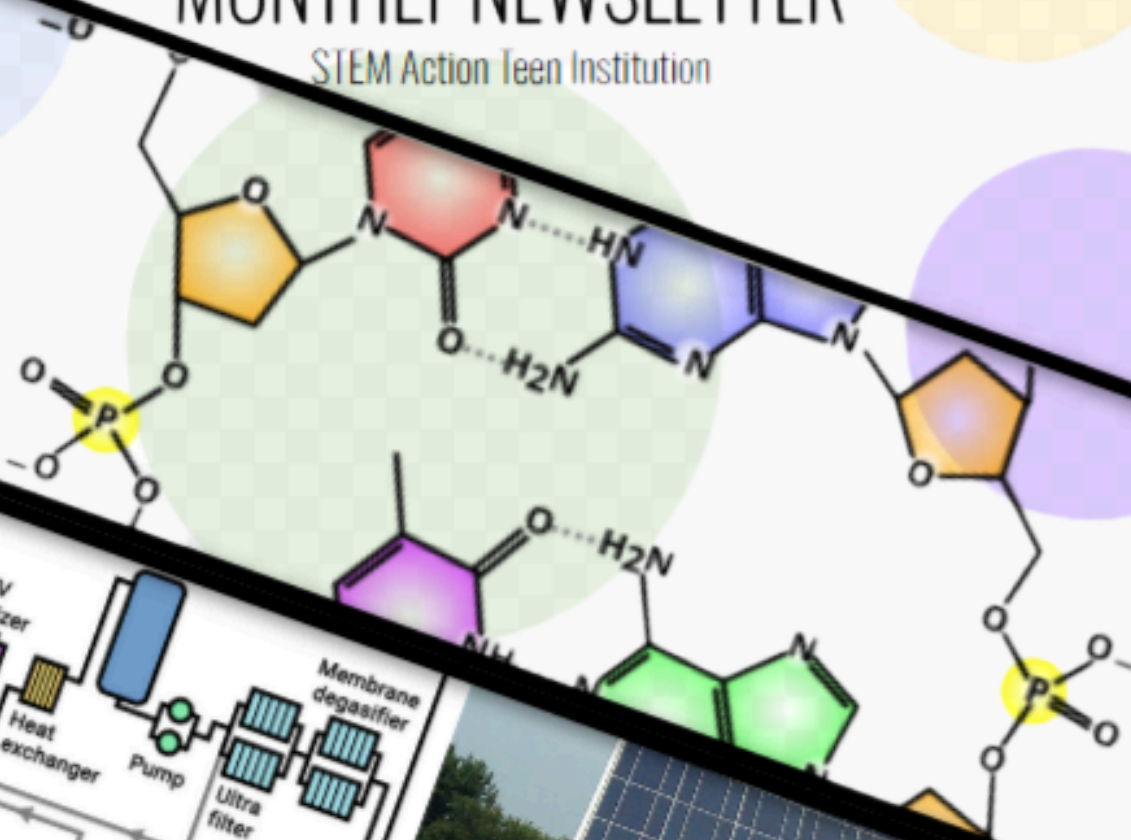
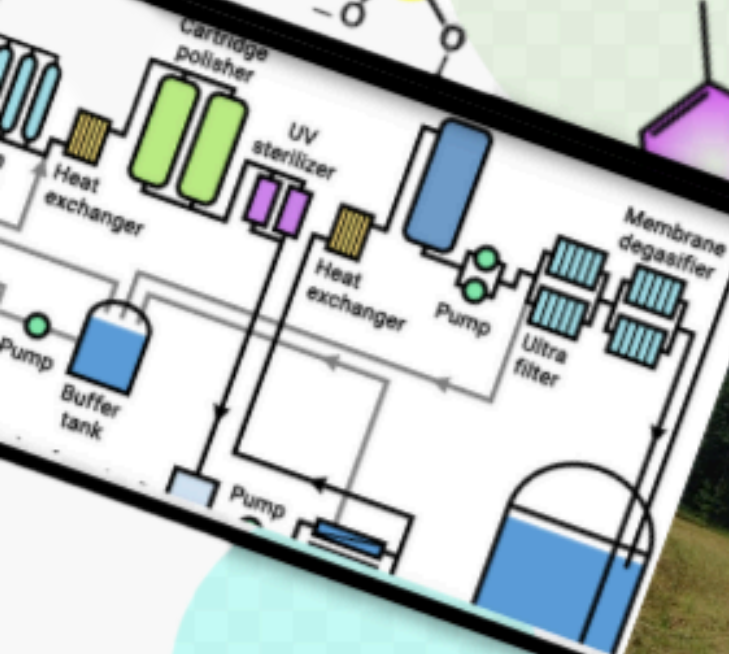
Seeking Science VOL 41

SEEKING SCIENCE

MONTHLY NEWSLETTER

STEM Action Teen Institution

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**The Chemistry Behind
Water Purification**
Brandon Wang

**DNA: The Code for
Your Body**
Brandon Pian

**The Future of
Renewable Energy**
Darren Wang

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The Connection Between Humor and Stress

Audrey Wu

The connection between humor and stress has interested scientists and psychologists for decades. In recent years, research has shed light on how humor can be a powerful tool for managing stress and promoting mental health. The relationship between humor and stress, the psychological and physiological benefits of humor, the mechanisms through which humor alleviates stress, and the role of humor in different aspects of life are all part of the connection between humor and stress.

Humor has been recognized as a natural and effective stress reliever. When people laugh, their brain releases endorphins, the body's natural feel-good chemicals, which can help reduce pain and elevate a person's mood. Laughter also decreases the levels of stress hormones such as cortisol and adrenaline, creating a sense of relaxation and well-being. Moreover, humor provides a mental break from the pressures and challenges of daily life, allowing people to momentarily distance themselves from their stressors and gain a fresh perspective. This mental shift can lead to improved problem-solving skills and increased resilience in the face of adversity.

The psychological benefits of humor extend beyond the immediate release of tension. Engaging in humor and laughter can foster social connections and strengthen relationships. When people share a laugh, it creates a sense of camaraderie and trust, which can be particularly valuable in stressful situations. Humor also enhances communication by breaking down barriers and facilitating open, honest conversations. For example, in a stressful workplace setting a light-hearted approach can diffuse conflicts, improve teamwork, and boost morale. As a result, individuals who regularly

incorporate humor into their interactions may find themselves better equipped to handle stress and maintain a positive outlook.

Humor's ability to alleviate stress is also rooted in its cognitive effects. When we encounter a humorous situation, our brain processes the incongruity between expectations and reality, leading to a moment of surprise and amusement. This cognitive shift can help reframe negative thoughts and reduce the perceived severity of stressors. Additionally, humor encourages creative thinking and problem-solving by promoting mental flexibility and divergent thinking. By approaching challenges with a playful mindset, individuals can discover innovative solutions and navigate stress with greater ease.

In conclusion, the connection between humor and stress is a fascinating interchange that underscores the importance of laughter in our lives. Humor not only provides immediate relief from stress but also offers long-term psychological and social benefits. By incorporating humor into daily routines and embracing a light-hearted perspective, people can enhance their resilience, improve their relationships, and maintain a healthier, more balanced approach to life. In a world that often feels overwhelming, humor remains a powerful and accessible tool for managing stress and fostering well-being.

The Future of Renewable Energy

Darren Wang

In the future, renewable energy will help solve many of the world's problems.

Renewable energy comes from natural sources like the sun, wind, and water. These sources never run out, and they are good for the planet. Unlike oil and coal, which cause pollution, renewable energy is clean and safe. If we use more renewable energy, we can protect the Earth and make it a better place to live.

One important type of renewable energy is solar power. Solar panels capture sunlight and turn it into electricity. Solar energy is great because the sun shines

every day, and it doesn't harm the environment. In the future, solar panels will be cheaper and better, so more people will be able to use them. Imagine if every home had solar panels! This would help power entire cities without causing pollution.

Another important renewable energy is wind power. Wind turbines, which are giant spinning blades, make electricity when the wind blows. Wind power is good for places with a lot of wind, like farms or the coast. In the future, we might have many wind turbines that can power entire communities. Wind power is also clean, which makes it great for the future.



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A solar cell panel in Hollis, New Hampshire by MarkBuckawicki from Wikimedia Commons

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Water is also a source of renewable energy. Hydropower uses moving water, like rivers, to create electricity. Dams or special machines use water to turn turbines and make power. Hydropower has been used for many years, and new technology might make it even better in the future. Scientists are working on ways to make sure hydropower is safe for animals like fish.

In conclusion, renewable energy will be very important in the future. With solar, wind, and water power, we can create clean energy for homes and businesses. As technology improves, renewable energy will become cheaper and easier to use. By using renewable energy, we can help protect the planet and make the future brighter for everyone.



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The Physics Behind Rainbows

Eddie Zhang

Rainbows are fascinating optical phenomena that result from the interaction of sunlight with water droplets in the atmosphere. The formation of a rainbow is governed by three key physical processes: refraction, reflection, and dispersion of light. These principles of physics explain how sunlight transforms into the vibrant arc of colors we see.

The process begins when sunlight enters a spherical water droplet in the atmosphere. As light moves from air (a less dense medium) into water (a denser medium), it slows down and bends, a phenomenon known as refraction. The angle and degree of bending depend on the wavelength of light; shorter wavelengths (blue and violet) bend more sharply than longer wavelengths (red and orange).

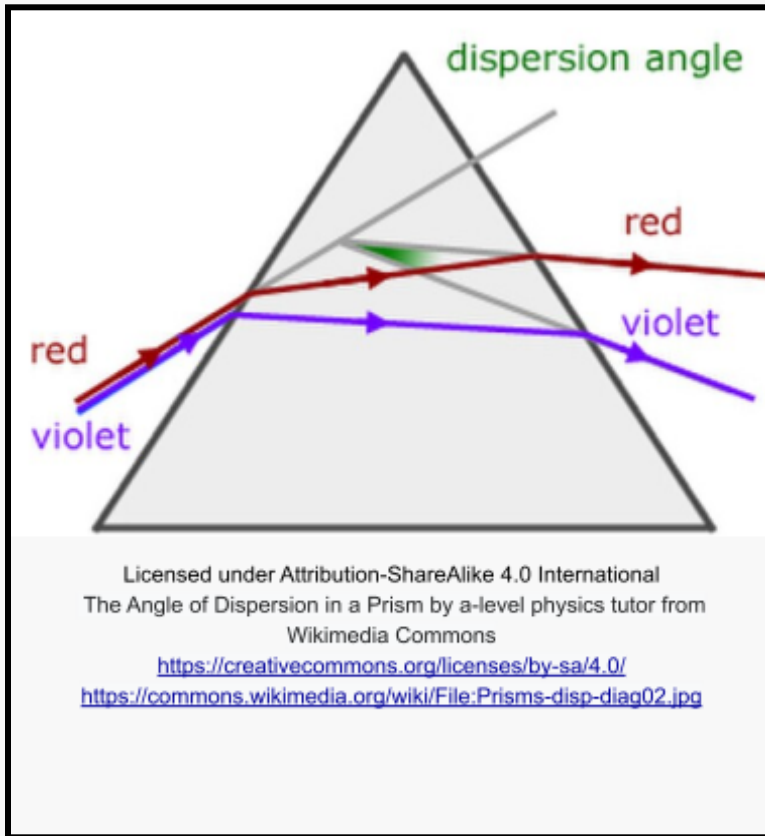
After refraction, the light travels through the droplet and strikes its inner surface. At this point, it undergoes internal reflection, bouncing off the inside surface of the droplet. The angle of incidence and the curvature of the droplet ensure that much of the reflected light remains within the droplet before exiting.

As light exits the droplet, it refracts again. During both refraction events, the light is dispersed into its component colors due to the wavelength-dependent bending. This dispersion separates white sunlight into its constituent spectrum: red, orange, yellow, green, blue, indigo, and violet (ROYGBIV). Each color emerges at a slightly different angle, with red light exiting at around 42 degrees and violet light at around 40 degrees relative to the incoming sunlight.

The combination of these processes within countless droplets creates a cone of light with the observer at its apex. Each observer sees their own unique rainbow because the light reaching their eyes comes from specific droplets that direct particular wavelengths toward them. The circular shape of the rainbow is a result of the consistent

angles at which the dispersed light emerges from the droplets. However, the ground often obscures the lower portion, making the rainbow appear as a semi-circle.

In rare cases, light can reflect multiple times inside the droplet before exiting. This



creates secondary rainbows, which are fainter and have reversed color order. Each additional reflection reduces the intensity of the light, making higher-order rainbows increasingly difficult to observe. In summary, rainbows are the result of sunlight undergoing refraction, internal reflection, and dispersion within water droplets. These interactions are rooted in the fundamental physics of light and its behavior in different media, demonstrating the natural

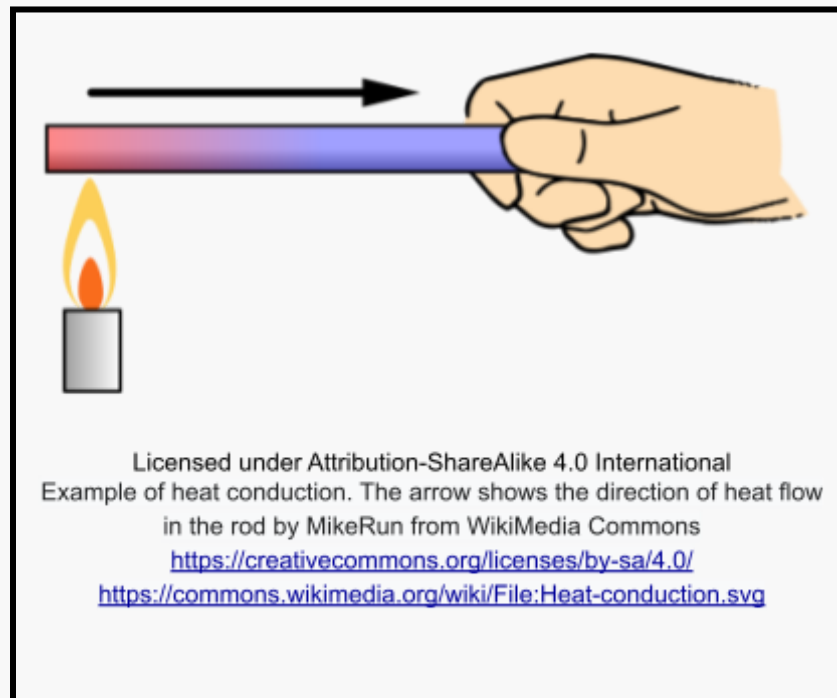
interplay between geometry and optical principles.

Thermodynamics

Mary Liang

To broadly define thermodynamics, it is the transfer of energy from one area to another. The base concept that everyone is familiar with is that heat is a form of energy, along with light, electricity, sound, etc. It hadn't been since 1798 that a British military engineer, Sir Benjamin Thompson recognized that the same heat is generated as the barrel of a fired cannon and turning a blunt tool repeatedly. Along with many other engineers and mathematicians, they developed their observations into what people would now call thermodynamics.

The Zeroth law of thermodynamics is that if two systems are in equilibrium with each other, and one of them is in equilibrium with a third system, then the first two systems are in equilibrium with the third system. The first law, or the law of conservation of thermodynamics states that energy cannot be destroyed nor created, but can be converted



into another form. The second law asserts that two systems work to form equilibrium. For example, when an ice cube is placed near a heated pan, it will start to melt since the pan releases heat into the air as the ice cube absorbs the heat and melts. It is also stated that

the energy wouldn't move back to the original owner, as hot gas mixed with cold gas would stay at its current temperature. The third law of thermodynamics illustrates that the entropy (measurement of the degree of randomness) of a crystal in its most solid form approaches zero as the temperature almost draws near absolute zero.

Thermodynamics can be found commonly in everyday life, like heating water, touching something with a different temperature, or the wiring of a computer fan. But for the people of the mid-1800s, heat was still a confusing topic. In 500 BC, some assumptions turned into facts in the modern-day world. It took humanity 2,000 years to get the concept of heat that we have today. The first man to discover thermodynamics was Sadi Carnot, who's titled the "father of thermodynamics" and also published a book on the concept of heat power, and engine efficiency. Through this stepping stool of thermodynamics, we might eventually find new concepts that could be crucial to the world of science.

The Chemistry Behind Water Purification

Brandon Wang

Water purification is a very important process to ensure that water is safe for drinking and free of agents injurious to health. This is a remarkable combination of physical and chemical methods through which impurities are removed and made fit for use in daily life.

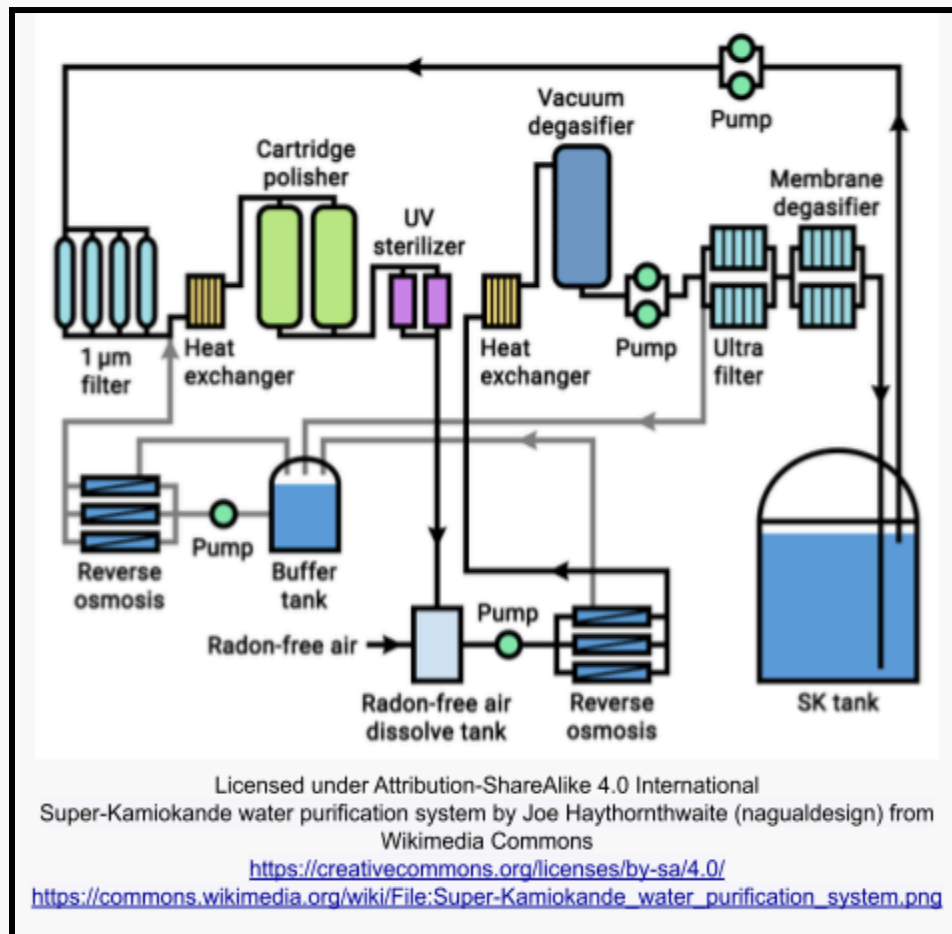
It begins with coagulation and flocculation, in which a chemical is added to the water-alum-aluminum sulfate, or iron III chloride is generally used. The chemical neutralizes the negative charges on dirt, bacteria, and other fine particles; once their negative charges are neutralized, particles begin clumping together and forming larger particles called "flocs." As the water proceeds from this, flocs become heavier, at which point gravity takes them down to the bottom of a tank at the sedimentation stage and removes most of the suspended solids.

Next, filtration allows water to pass through layers of sand, gravel, and activated carbon. With these layers, most of the smaller particles remain behind, including some microorganisms, while the activated carbon absorbs chemical impurities, improving both the taste and smell of the water.

Then, disinfection of the filtered water is carried out to eliminate the potentially harmful microorganisms, such as bacteria and viruses. Chlorine is widely used since this gas forms a very powerful disinfectant called hypochlorous acid upon reacting with water. Other methods can involve treatment with ozone or even ultraviolet light. For example, UV light works on the principle of damaging the DNA of pathogens, thus rendering them non-viable.

In some cases, with highly saline or unusually contaminated water, further steps such as reverse osmosis or desalination must also be undertaken. These processes force water through special membranes that remove salts and impurities. Finally, chemicals such as lime or soda ash are added to adjust the pH of the water to make it safe to drink and non corrosive to pipes.

All these combined render clean and safe water, proof that chemistry is vital in ensuring public health. Beyond drinking water, purification enables agriculture, industry, and the health of ecosystems. As freshwater becomes a growing concern, future advances in purification technology will help ensure that this life necessity continues to be available for years to come.



DNA: The Code for Your Body

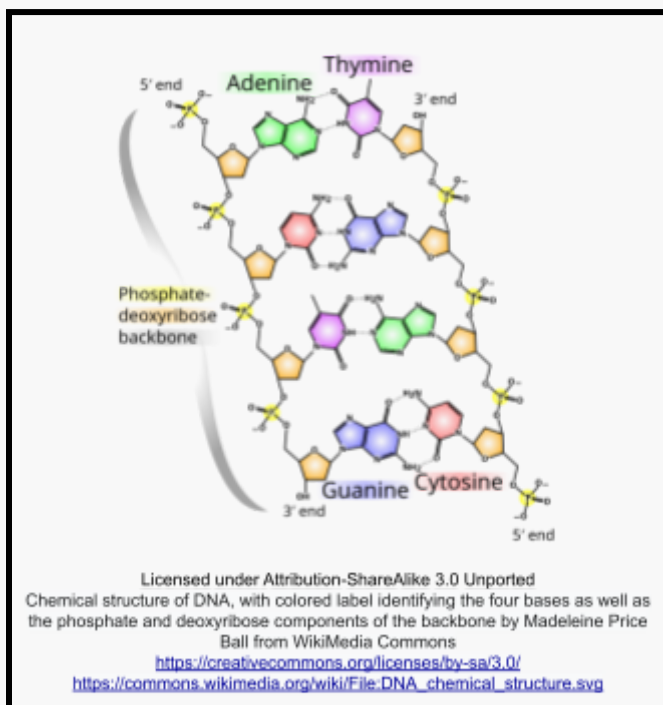
Brandon Pian

DNA, or deoxyribonucleic acid, is the blueprint of life, containing genetic instructions for growth, development, and reproduction in all living organisms. In 1953, James Watson and Francis Crick discovered its double-helix structure, which transformed the field of genetics and deepened our understanding of life.

DNA consists of smaller units called nucleotides, each nucleotide is made up of a 5 carbon sugar, a phosphate group, and a nitrogenous base. There are four bases: adenine (A), thymine (T), cytosine (C), and guanine (G). These bases pair in a specific way (A with T, C with G) to form the "rungs" of the DNA ladder. The sequence of these bases acts like a biological code, determining an organism's traits and directing cellular

functions. These bonds are held together by hydrogen bonds with 2 hydrogen bonds between adenine and thymine and 3 hydrogen bonds between guanine and cytosine.

One of DNA's most vital functions is replication. Before a cell divides, during S phase DNA unwinds and each strand serves as a template for a new complementary strand. This ensures that genetic information is passed down accurately. Additionally, DNA directs the



production of proteins through transcription and translation, processes essential for sustaining life.

DNA replication occurs when helicase unwinds the double helix, and the primer marks areas for the DNA polymerase to start building new strands by pairing nucleotides (A with T, C with G). The leading strand is copied continuously, while the lagging strand is made in fragments joined by ligase. This process ensures each new DNA molecule has one original and one new strand.

As scientists continue to explore DNA's potential, its role in shaping the future of biology, medicine, and technology remains profound. DNA truly is the foundation of life itself.

ChatGPT vs DeepSeek

Richard Wang

Recently, at the start of 2025, a Chinese artificial intelligence model, DeepSeek, has shot to the top of app store downloads. In addition, it was built at less than a tenth of the price of OpenAI's ChatGpt. This sudden appearance caused chip-making company, Nvidia to lose approximately \$600 billion of its market value in a day. Companies like Nvidia produce the chips to run the very demanding computations for these AI models. Currently, DeepSeek's R1 model is being compared to ChatGPT. Each of these models has its own advantages over the other and appeals to different audiences.

DeepSeek contains many advantages that cater to specific needs that may be different from ChatGPT's. DeepSeek is a technical model that is superior in logic and mathematical tasks. It is very affordable as its API access starts at only \$0.14 per million tokens compared to ChatGPT's \$7.50 per million token usage, making it good for developers on a budget. DeepSeek, unlike ChatGPT, is open source meaning users can



run the model locally which gives them greater control over their data. This appeals to people who are conscious about their privacy. DeepSeek is a model that is superior in technical problem-solving, logic, analysis, and math problems. It is great and faster at providing task-oriented responses.

ChatGPT, however, still beats the recent DeepSeek in some categories. ChatGPT excels and is

better than DeepSeek at conversational output. It is great at creating a natural and casual tone in its responses. It is a great option if the user needs a chatbot for writing, storytelling, brainstorming, or travel planning. It can also be used for simple and day-to-day questions. Also, ChatGPT in general is faster than DeepSeek in long and multi-part queries. With a human-friendly approach, ChatGPT has more user-friendly customization options making it appealing to a larger crowd of consumers.

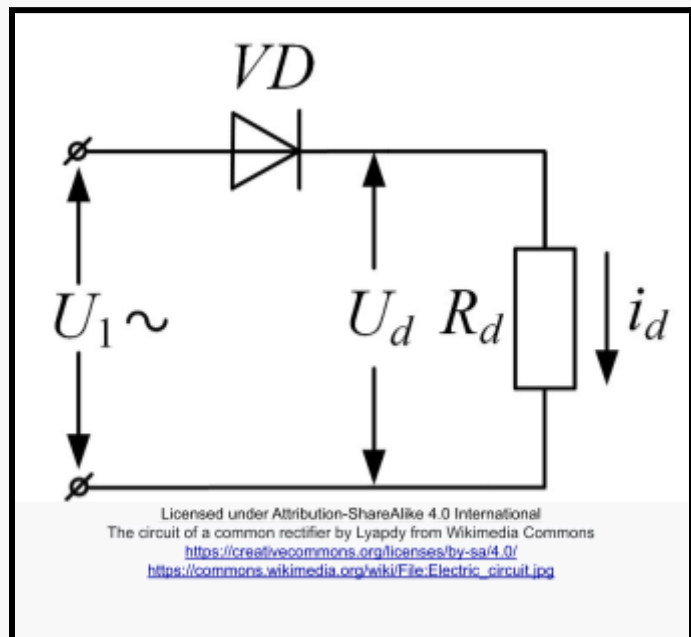
ChatGPT and DeepSeek's models offer distinct strengths that cater to distinctive groups. DeepSeek shines as it is more cost efficient and is better at technical prompts such as coding or mathematics. ChatGPT is able to compete with its adaptability, humanistic replies, and user-friendly design.

The Significance of Electric Potential

Kenny Wu

Electric potential is a fundamental concept in the world of Physics, it reflects the potential energy per unit charge at a point in an electric field. Electric potential is measured in volts (V) and stands for the work needed to move a unit charge from a reference point, which tends to be set at infinity to have an easier time dealing with calculation. The concept of electric potential serves a prominent role in sorting out the logic behind energy transfer in electrical circuits, capacitors, and even biological operators.

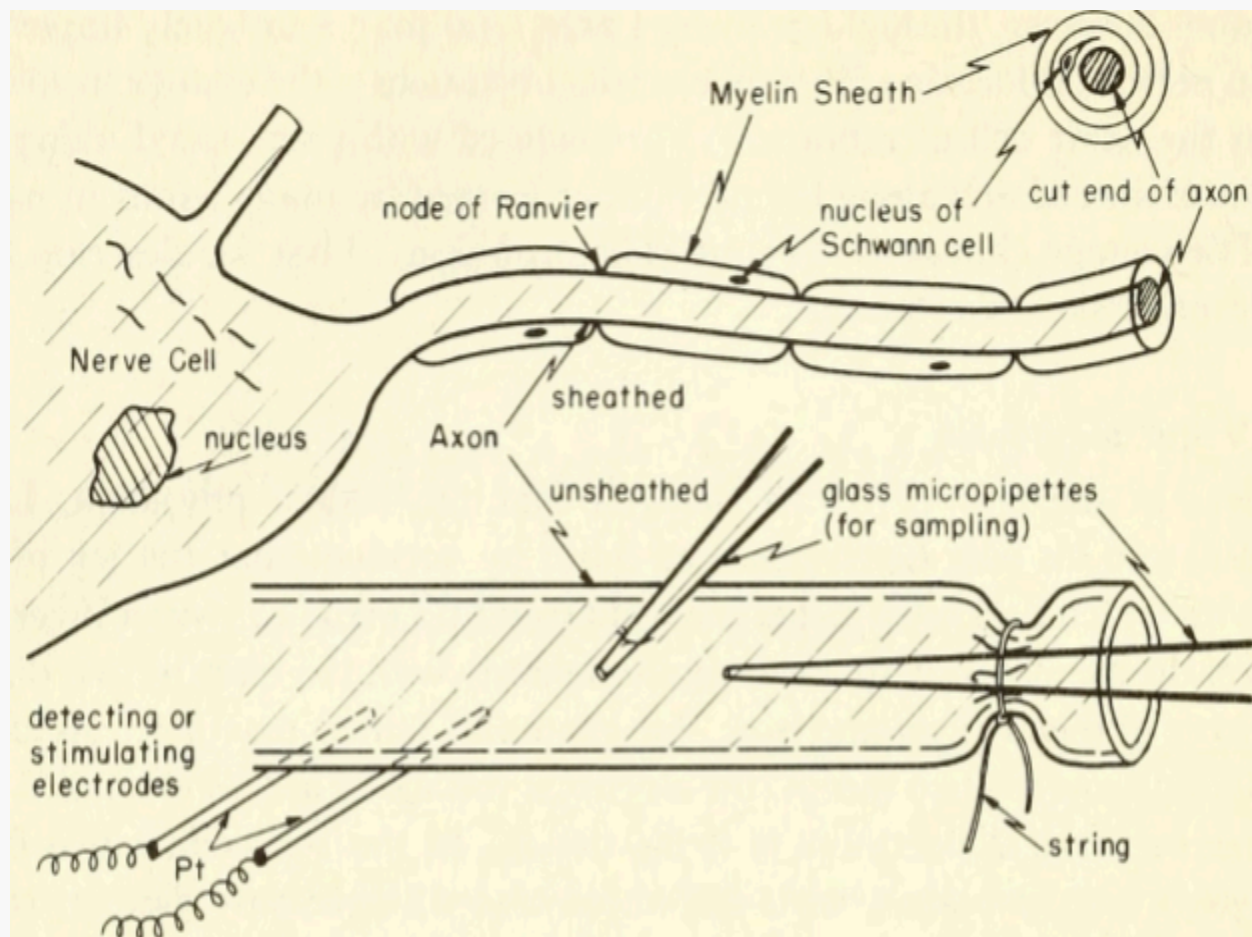
One most notable application of electric potential is in electrical circuits, where a voltage reservoir forms a difference in electrical potential between two points. The potential difference created results in the movement of charges, often referred to as electric current. A higher electric potential difference would ensure a greater abundance of energy to perform work, such as lighting a bulb or powering a motor.



Electric potential also rules the behavior of capacitors, storing energy in an electric field between two conductive plates. When a capacitor is connected to a voltage source, it forms an electric field by ramping up charges on the two plates. The stored energy could be released in quick succession, reflected in applications like camera flashes and power backup systems.

Beyond electronics, electric potential is utilized substantially for biological means. Nerve impulses in the human body heavily rely on changes in electric potential across cell membranes. The constant changes are referred to as action potentials, granting our options to think, move, and react by allowing communication between neurons.

All in all, electric potential is a concept that lies beyond the theoretical concept and acts as the foundation of energy transfer in electrical and biological systems. Understanding how potential differences influence charge movement helps engineers design efficient circuits and provides insight into essential biological functions.



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Figure 10-1 shows nerve cells and axons, highlighting the squid axons role in studying nerve conduction by Casey and Edmund Jeremiah from Flickr

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The NP Problem: One of Mathematics' Greatest Mysteries

Owen Chen

In the world of mathematics and computer science, some problems are easy to solve, while others seem impossible to compute efficiently. From planning delivery routes to securing online transactions, many real-world challenges rely on solving complex problems quickly. At the heart of these challenges lies one of the biggest unanswered questions in theoretical computer science: the P vs. NP problem. This problem asks whether every problem whose solution can be verified quickly can also be solved quickly. If $P = NP$, countless currently difficult problems would become easy to compute, revolutionizing fields such as cryptography, optimization, and artificial intelligence. This essay explores the nature of NP problems, their significance, famous examples, and the broader implications of solving the P vs. NP question.

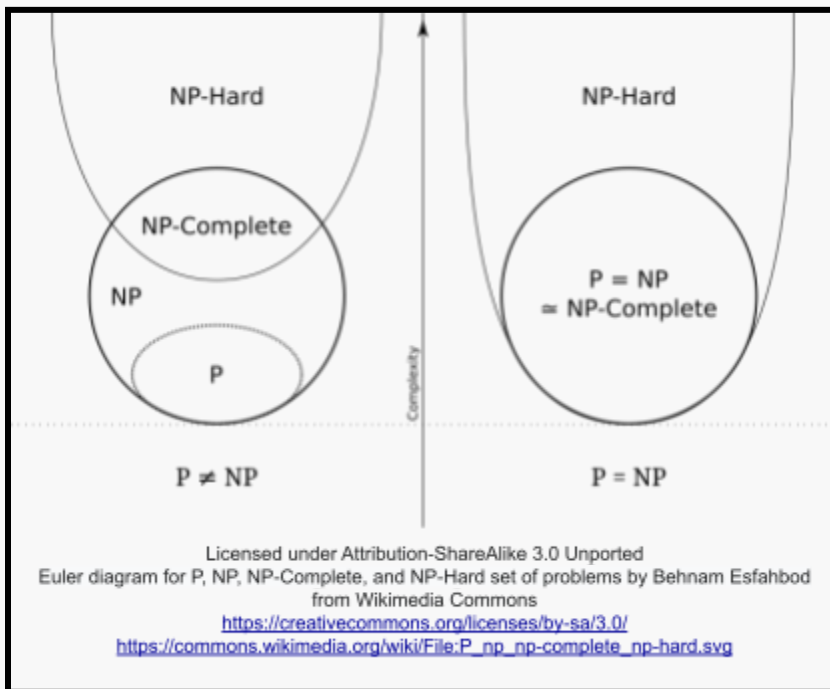
To understand the NP problem, we must first define P and NP. The class P (polynomial time) consists of problems that can be solved efficiently—meaning their solutions can be found in polynomial time relative to the input size. For example, sorting a list of numbers or finding the shortest path in a graph can be done in polynomial time using known algorithms. The class NP (nondeterministic polynomial time) consists of problems for which a given solution can be verified quickly, but finding the solution may not necessarily be easy. A classic example is Sudoku: if someone provides a solution, it is easy to check whether it is correct, but finding that solution from scratch is much harder.

Within NP, a special category called NP-complete exists. These are the hardest problems in NP—if any NP-complete problem can be solved in polynomial time, then all NP problems can be solved in polynomial time, proving $P = NP$. The question of whether

P equals NP remains one of the most profound open problems in mathematics.

NP-complete problems appear in various fields, from logistics to artificial intelligence. Some well-known examples include the Traveling Salesman Problem (TSP), which asks for the shortest possible route that visits each city exactly once and returns to the starting point. This problem is crucial in transportation and logistics but remains computationally difficult to solve. The Boolean Satisfiability Problem (SAT) was the first problem proven to be NP-complete and involves determining whether there exists an

assignment of true/false values to variables in a logical formula that makes the entire formula true. Another example is the Graph Coloring Problem, which seeks to assign colors to graph vertices so that no two adjacent vertices share the same color, with applications in scheduling and frequency assignment. The Knapsack Problem is another widely studied example, where the goal is to select a subset of



items that maximizes value while staying within a weight limit, a problem relevant to resource allocation and finance.

NP problems are not just theoretical—they play a fundamental role in modern computing and everyday life. Cryptography relies on NP problems being difficult to solve, with modern encryption methods such as RSA and ECC depending on their complexity. If $P = NP$, these encryption methods would be easily breakable, endangering secure communication worldwide. Optimization problems in logistics, scheduling, and

manufacturing often involve NP-hard problems. The ability to solve these efficiently would dramatically improve operations across industries. In artificial intelligence and machine learning, many AI challenges, such as training neural networks and game strategy computations, involve NP-complete problems. If $P = NP$, AI could achieve breakthroughs that are currently infeasible.

The P vs. NP problem remains unsolved, with profound implications for both theory and practice. If $P = NP$, every problem with a quickly verifiable solution could also be solved efficiently, leading to groundbreaking advances in AI, medical research, and automation, but also compromising cryptographic security. If $P \neq NP$, many problems will remain inherently hard to solve, but cryptographic security would remain intact, and computational complexity theory would be reinforced. Mathematicians and computer scientists have worked on this problem for decades, but no definitive proof has been found. The Clay Mathematics Institute has even offered a \$1 million prize for a correct proof, demonstrating its importance.

The NP problem is one of the most fundamental and intriguing questions in computer science and mathematics. Understanding whether $P = NP$ could change how we solve problems in cryptography, optimization, and artificial intelligence. While the answer remains elusive, its implications reach far beyond academia—potentially reshaping technology and society. As research continues, the pursuit of a solution remains one of the greatest intellectual challenges of our time.

Natural Selection

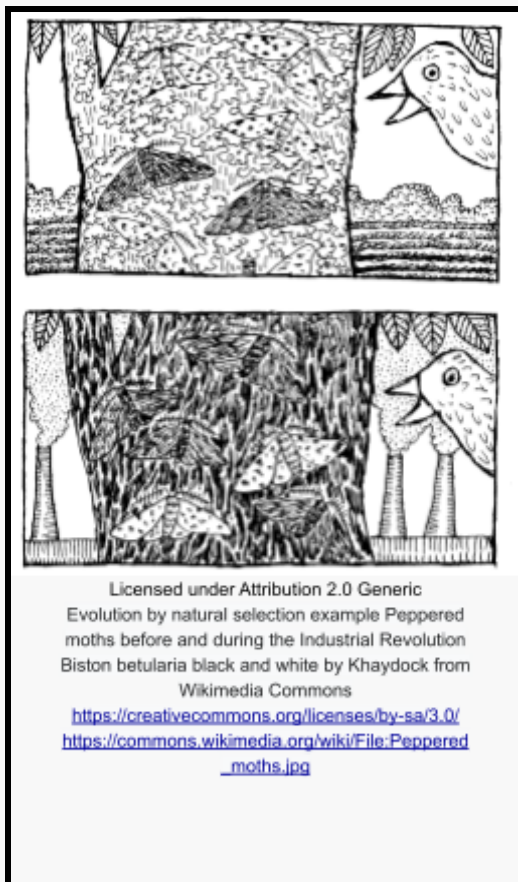
Ethan Liu

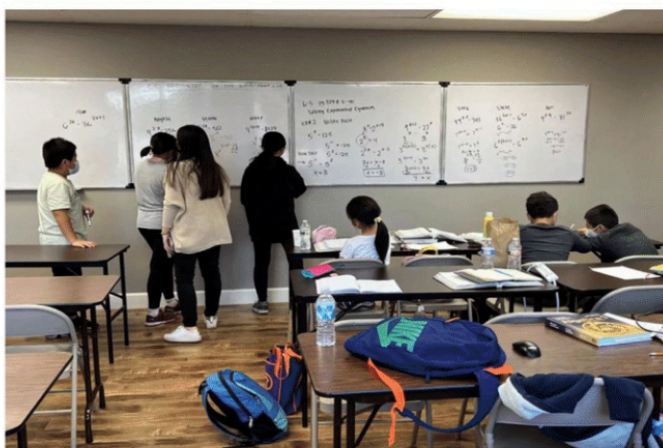
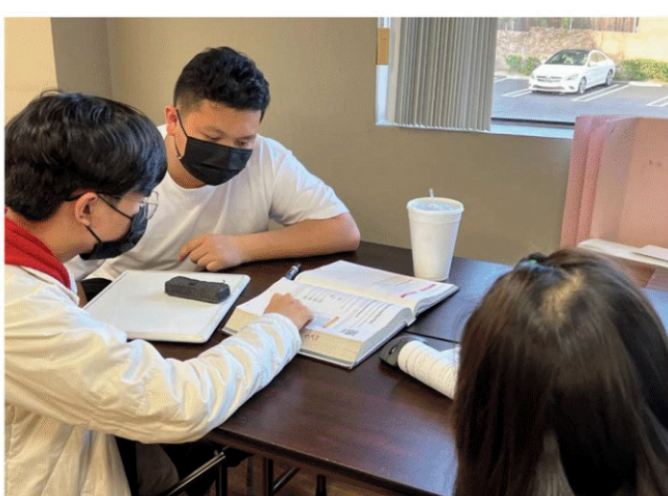
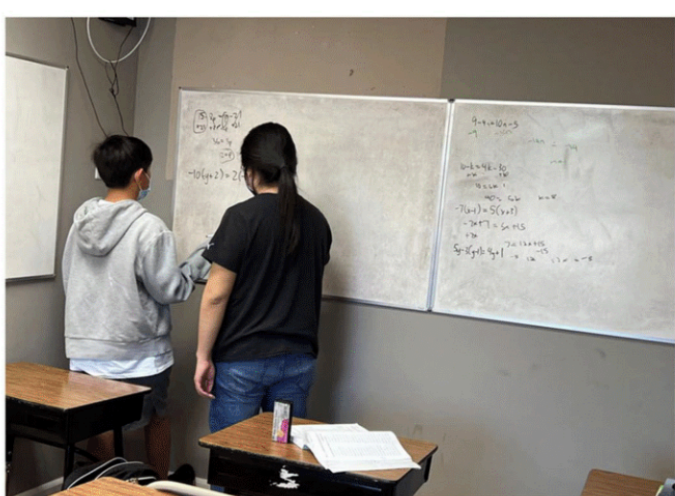
Natural selection, a key mechanism of evolution, explains how species adapt and survive in their environments over time. Proposed by Charles Darwin, this process occurs when organisms with traits better suited to their surroundings are more likely to survive and reproduce, passing on those advantageous traits. Over generations, these traits become more common, gradually shaping the population to be better adapted to its environment.

For example, the famous case of peppered moths during the Industrial Revolution illustrates natural selection in action. Before the revolution, most moths were

light-colored, blending in with lichen-covered trees. As pollution darkened the trees, darker moths became less visible to predators and survived more often than their light-colored counterparts. Over time, the moth population shifted to predominantly dark-colored moths, demonstrating how environmental changes can drive evolution.

Natural selection is essential to understanding biodiversity and how species change over time. It highlights the interconnectedness of organisms and their environments, showing how survival and reproduction shape life on Earth. As environments continue to change, natural selection remains a driving force in evolution, ensuring that life adapts to new challenges.





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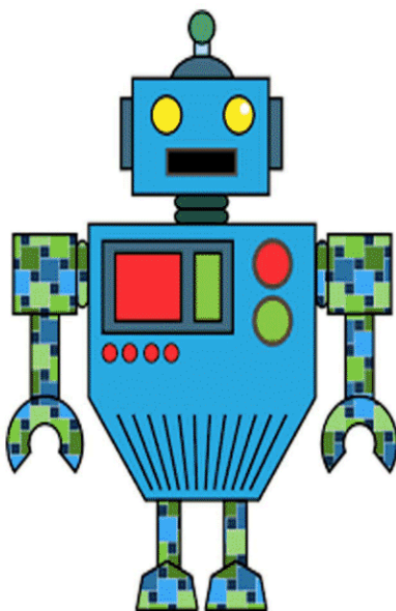
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